

Intro SQL Module

[Databases
Keys
CRUD
Aggregate Queries
Sub Queries
Indexing
Transaction
Schema Design

12 Classes

[MySQL Server
MySQL Workbench
Salci's Database

Nikhil Jain

3+ years in Scaled

- ↳ SQL
- ↳ LLD
- ↳ LLD Projects

Principal Engineer @ Atlassian

- ↳ Conf DL
- ↳ Dev Productivity

3 years

AI PR Reviewer for BitBucket

SDE3 Amazon

- ↳ Retail Pricing System

2 years

Staff Engineer @ Walmart

- ↳ Competitive Matching

3 years

SDE2 @ Microsoft

- ↳ Bing Ads

2.5 years

SDE @ Amazon
↳ Imaging Systems 2 years

ASDE @ Mixys (finstar)

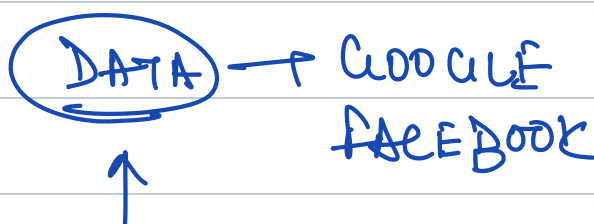
2014 MNIT Jaipur CSE

Root Cause Analyst

Expectations

- ① 9:10 → 10:00 ¹⁰ 10:10 → 10:45
 - ② Spoon feeding
-

"DATA IS KING"



OPENAI

Person → Grocery Shopping

 wallet

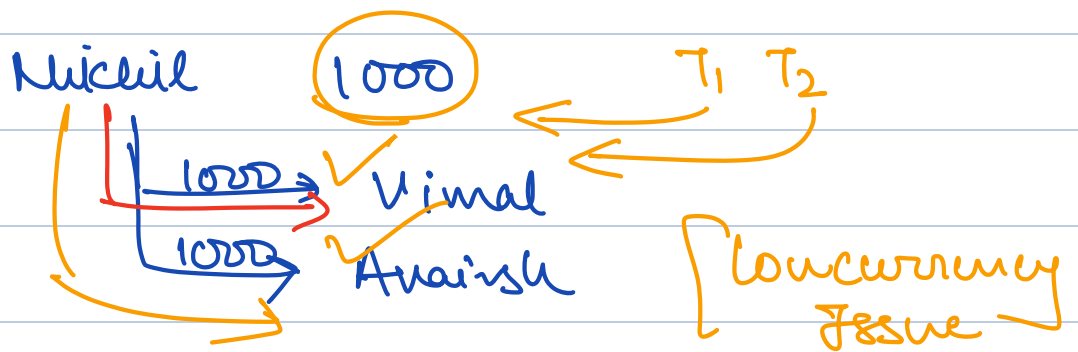
List of Henry → Phone
↳ Paper] Data

1 → BigBasket Paper

↳ EXCEL

4 to 5 kcal

➔	Micheil	SQLI	Batch A
	Micheil	LLD	Batch B A



Database

↳ Collection of Related Data

Scaler

- ↳ user
- ↳ Students
- ↳ Instructors
- ↳ Classes
- ↳ Butcher

facebook

- ↳ user
- ↳ Posts
- ↳ Comments

DBMS

↳ Database Management System

Software to manage Data

↳ Operations → Create

Read

Update

Delete

CRUD

ID Name PSP Attendance

DB

↳ Relational DB

↳ Non-Relational DB

Relational DB

↳ Student Profile → Name, Phone, Email
↳ Student Class Data → PsP, Attendance

Non Relational DB

↳ Survey form] Document

Key Value Pair

Config ↗ Map (String String)
URL - x
Password - y
Thread Count = 2

Properties of RDBMS

① Every Row should be Unique

② Data will stored in Tables

③ Data in same col will be of same type

psp → int

↳ 0 1 2 - -

Zero One Two

④ Values are Atomic

List <Phone> → Array → same it
↳ Atomic

⑤ Row | Col seq. is NOT Guaranteed

*

⑥ Name of col is always unique

Keys

Why do we want Rows to be Unique

- Relations to be Unique
- Avoid Duplication

Primary Key

Foreign Key

Class

Class Id

Class Inst Id

foreign key

Super Key

Candidate Key

g

User

Name

Email

Phone

psp

Email + Phone — UNIQUE

Name + EMAIL + PHONE —

Name + Email + Phone + Psp —

↳ Super Keys

↳ Combination of cols that can uniquely identify a Row

Candidate Key

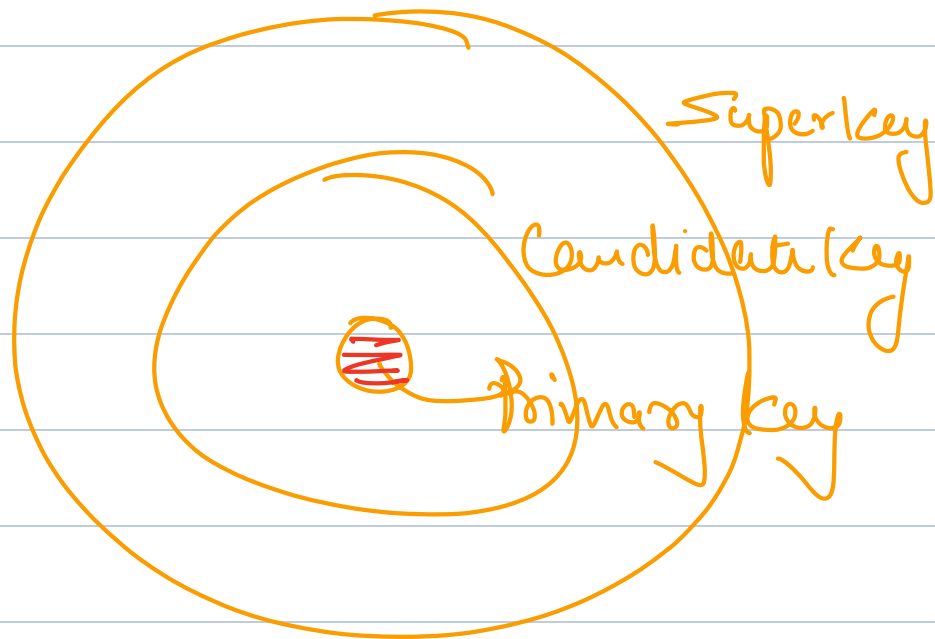
~~Name + Email + Phone~~

~~EMAIL + Phone~~] Candidate Key

ID Name Email Phone Psp

Primary Key	ID	—	SUPER Key	Candidate Key
	Name	—	X	X
	EMAIL	—	X	X

Phone	—	x	x
<u>Email + Phone</u>	—	✓	✓
Name + Email + Phone	—	✓	x
Id + Email + Phone	—	✓	x



Composite key → Combination of any 2 cols

Name + PSP

Order

<u>Order-id</u>	<u>User-id</u>	<u>Product-id</u>	<u>Qty</u>	<u>Payment-id</u>	<u>TS</u>
1	1	1	1	1	359
1	1	2	1	1	359
2	2	1	2	2	370
3	1	1	3	3	371

Name + PSP → Score